

GOKUL CHANDRASEKARAN

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PROFESSIONAL SUMMARY

Mechanical engineer specializing in thermal-fluid systems with 7+ years of experience in two-phase cooling, porous media, and energy systems. Experience spans federally funded research (DARPA, NSF, NASA) and industry R&D, with contributions resulting in peer-reviewed publications and a U.S. provisional patent. Active in the thermal engineering community through conference presentations, peer review, and professional service.

CORE EXPERTISE

Thermal & Fluid Systems: Two-phase cooling, heat pipes, porous media, atmospheric water harvesting

Experimental Methods: Thermal diagnostics, instrumentation, high-temperature and vibration testing

Modeling & Simulation: COMSOL Multiphysics, ANSYS Fluent, MATLAB

Data & Systems: LabVIEW automation, experimental design, uncertainty analysis

Manufacturing: Additive manufacturing (metal/polymer), wick structures, composite materials

PROFESSIONAL EXPERIENCE

Lead, Thermal Engineering

Global Cooling Technology Group, LLC – Tempe, AZ | Jan 2026 – Present

- Lead experimental evaluation of closed-loop pulsating heat pipe (CLPHP) systems for high-power electronic cooling
- Define and execute thermal performance testing using automated experimental platforms
- Review experimental methodologies and datasets to ensure consistency and accuracy
- Contribute to DARPA SBIR Phase II program activities, including technical reviews under vibration and elevated temperature conditions
- Support dissemination of thermal management research through conference publications

Postdoctoral Research Scholar

Arizona State University – Tempe, AZ | Feb 2026 – Present

- Conduct experimental research on coated desiccant systems for atmospheric water harvesting
- Design and operate controlled testbeds for performance evaluation across environmental conditions
- Support proposal development and collaborative research in thermal-fluid systems

Graduate Research Assistant (PhD)

Arizona State University – Tempe, AZ | Aug 2018 – Dec 2025

- Conducted experimental research in atmospheric water harvesting, two-phase cooling, and porous media systems
- Developed ultrasound-enhanced desiccant regeneration method achieving **10–25% reduction in energy consumption**, resulting in a publication and U.S. provisional patent
- Performed experimental and modeling studies on metal-coated polymer foams
- Contributed to NSF STTR Phase I (2124797) and NSF SBIR Phase II (2321862) projects; identified as a contributing participant in NSF reporting
- Supported NASA ESI project (80NSSC22K0260) through experimental validation of additively manufactured wick structures
- Designed permeability measurement systems for additively manufactured wick structures
- Developed automated thermal testing platforms (LabVIEW, MATLAB)
- Built experimental setups for testing under elevated temperature and vibration conditions
- Mentored graduate researchers in experimental methods and data analysis

Thermal Modeling Engineering Intern

Global Health Labs – Bellevue, WA | May 2023 – Aug 2023

- Developed COMSOL-based thermal models and integrated simulation with experimental data to support design refinement
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KEY CONTRIBUTIONS & IMPACT

- Developed ultrasound-assisted desiccant regeneration method reducing energy requirements by up to 25%, resulting in patent filing and publication
- Built experimental systems for evaluating two-phase cooling technologies under federally funded and industry programs
- Disseminated research across academic, industry, and defense-oriented venues (TFEC, AZ Water Conference, GOMACTech)
- Contributed to atmospheric water harvesting research presented at international conferences, including IAWHA summits
- Designed experimental methods for characterizing permeability and thermal performance of porous and additively manufactured materials

PUBLICATIONS & INTELLECTUAL PROPERTY

Patent: Ultrasound-Enhanced Desorption for Atmospheric Water Capture, U.S. Provisional Patent #63/812,456 (2025)

Selected Publications:

- Rafat et al., *Environments*, 2025
- Alenezi et al., *International Journal of Heat and Mass Transfer*, 2023
- Dai et al., *International Journal of Thermal Sciences*, 2021

CONFERENCE CONTRIBUTIONS & PROFESSIONAL RECOGNITION

- ASTFE TFEC 2026 – Oral presentation: *Desorption Process Enhancement for Atmospheric Water Harvesting Using Ultrasound*
- AZ Water Conference 2026 – Selected speaker, 99th Annual Conference
- ASTFE TFEC 2025 – Poster presentation; NSF Travel Award recipient
- GOMACTech 2026 – Co-author; work presented under DARPA SBIR Phase II and NSF SBIR/STTR programs
- ASTFE Student Committee Member (2025–2026) – Conference programming and student engagement

FEDERALLY FUNDED RESEARCH CONTRIBUTIONS

- DARPA SBIR Phase II – Pulsating heat pipe cooling systems (Contract No: 140D0425C0005, awarded to GCTG)
- NSF SBIR Phase II – Two-phase cooling technologies (Award No: 2321862)
- NSF STTR Phase I – Pulsating heat pipe systems (Award No: 2124797)
- NASA Early Stage Innovations (ESI) – Additively manufactured heat pipe radiators (Award No: 80NSSC22K0260)

PEER REVIEW & PROFESSIONAL SERVICE

Reviewer: ASME Journal of Thermal Science and Engineering Applications; IEEE Electron Devices Magazine; IEEE T-CPMT; ITherm Conference; Ain Shams Engineering Journal

Memberships: ASME; ASTFE; IEEE Electronics Packaging Society

MEDIA & PUBLIC ENGAGEMENT

- Research featured in *3D Printing Industry*, *VoxelMatters*, and *ASU News* on 3D-printed thermal technologies
- Featured in innovation spotlight by The Engine (MIT) and NSF Engines ecosystem

EDUCATION

PhD, Mechanical Engineering – Arizona State University, 2025

MS, Mechanical Engineering – Arizona State University, 2018

B.Tech, Mechanical Engineering – IIT Madras, 2016

AWARDS

- NSF Travel Award – ASTFE TFEC 2025
- NSF I-Corps Aspire & Lab2Launch – Technical Lead (QuantumAtmos Team)
- ITherm Heat Sink Design Challenge 2019 – Finalist
- SpaceX Hyperloop Competition 2017– Finalist